Introduction to Programming: Quiz 1

Date: September 17, 2026 **Time:** 1700 – 1745 hrs **Marks:** 10 **Weightage:** 10%

1. (2 marks) Write a recursive function `dropOdds :: [a] → [a]` that drops the elements at odd indices of a list (indexing from 0). For example:

```
dropOdds [1,2,3,4,5] = [1,3,5]
```

2. (2 marks) Write a function `sums` that takes a list of lists of numbers `xs` and returns the list of sums of the non-empty lists in `xs`. For example:

```
sums [[1..5], [], [2,6]] = [15,8]
```

3. (2 marks) Consider the function

```
f xs = foldr (/=) False xs
```

Find the type of `f`, and compute `f` on two example inputs.

4. (4 marks) Given the decimal expansions of two numbers (as lists of digits of equal length, most significant digit first), compute the decimal expansion of their sum. For example:

```
add [0,8,5,3,3] [0,2,8,6,8] = [1,1,4,0,1]
```